Experiment 14: OFDM Delay-Profile Estimation

Using Sionna RT at 5.9 GHz

V2X–ISAC Research

November 1, 2026

Abstract

This experiment investigates OFDM-based delay-profile estimation using a multipath channel generated through NVIDIA Sionna RT.

The simulation uses a carrier frequency of 5.9 GHz, a sampling frequency of 100 MHz, and an FFT size of 1024. Twelve valid ray entries are obtained from the Sionna RT propagation simulation and converted into a discrete-time channel using fractional-delay interpolation.

A QPSK OFDM signal is transmitted through the discrete channel. The estimated frequency response is converted into the delay domain using an inverse FFT. The resulting delay profile is compared with the reference channel and the propagation delays obtained from ray tracing.

The peak detection algorithm identifies five prominent delay-domain peaks from the twelve valid ray entries. The relative reconstruction RMSE is approximately $1.84 \times 10^{-}16, demonstratingnumericalconsistencybetweentheestimatedandreferencediscretechannels$ $freeconditions$.

Contents

1 Introduction 4

2 Simulation Configuration 4

3 Mathematical Methodology 5
- 3.1 Multipath channel representation . . . 5
- 3.2 Discrete-time channel . . . 5
- 3.3 OFDM frequency-domain channel estimation . . . 6
- 3.4 Conversion to the delay domain . . . 6
- 3.5 Magnitude normalization . . . 7
- 3.6 Local peak detection . . . 7

4 Ray-Tracing Results 7

5 Experimental Results 8
- 5.1 Detected delay-domain peaks . . . 8
- 5.2 Graphical results . . . 8
- 5.3 Description of the results . . . 8

6 Comparison of Ray Delays and Detected Peaks 9

7 Reconstruction Validation 9

8 Discussion and Research Findings 10

9 Limitations 11

10 Relevance to V2X-ISAC Research 11

11 Conclusion 12

1 Introduction

The objective of this experiment is to investigate the delay-domain characteristics of a multipath wireless channel using OFDM signal processing.

In the previous experiment, OFDM channel estimation was validated in the frequency domain. In this experiment, the estimated frequency response is transformed into the delay domain using the Inverse Fast Fourier Transform (IFFT).

The main objectives are:

1. Convert the estimated frequency response into a delay-domain representation.
2. Identify prominent delay-domain peaks.
3. Compare detected peaks with Sionna RT propagation delays.
4. Investigate the effect of finite bandwidth on the identification of closely spaced paths.
5. Establish a baseline for subsequent V2X-ISAC sensing experiments.

2 Simulation Configuration

The simulation uses the Sionna RT `simple_street_canyon` environment.

The transmitter and receiver positions are:

$$\mathbf{p}_{TX} = [-20, 0, 2]^T \text{ m} \tag{1}$$

$$\mathbf{p}_{RX} = [20, 0, 2]^T \text{ m} \tag{2}$$

The separation between them is:

$$d_{TX,RX} = 40 \text{ m} \tag{3}$$

The simulation parameters are listed in Table 1.

Table 1: Experiment 14 simulation parameters

Parameter	Value
Carrier frequency	5.9 GHz
Sampling frequency	100 MHz
FFT size	1024
Modulation	QPSK
Cyclic prefix	128 samples
Subcarrier spacing	97.65625 kHz
Useful symbol duration	10.24 microseconds
Channel length	60 samples
Valid ray entries	12
Noise	Disabled
Peak threshold	-25 dB

3 Mathematical Methodology

3.1 Multipath channel representation

The continuous-time multipath channel is represented by:

$$h(t) = \sum_{\ell=0}^{L-1} a_\ell \delta(t - \tau_\ell) \tag{4}$$

where:

- L is the number of propagation paths.
- a_ℓ is the complex path coefficient.
- τ_ℓ is the propagation delay.
- $\delta(t)$ is the Dirac delta function.

The Sionna RT simulation produced twelve valid ray entries:

$$L = 12 \tag{5}$$

The earliest and latest propagation delays were:

$$\tau_{\min} = 133.426 \text{ ns} \tag{6}$$

$$\tau_{\max} = 283.282 \text{ ns} \tag{7}$$

Therefore, the delay span is:

$$\begin{aligned} \Delta\tau &= \tau_{\max} - \tau_{\min} && (8)\\ &= 283.282 - 133.426 && (9)\\ &= 149.856 \text{ ns} && (10) \end{aligned}$$

3.2 Discrete-time channel

The sampling interval is:

$$T_s = \frac{1}{f_s} \tag{11}$$

For a sampling frequency of 100 MHz:

$$T_s = \frac{1}{100 \times 10^6} = 10 \text{ ns} \tag{12}$$

The fractional sample index of each propagation path is:

$$\nu_\ell = \tau_\ell f_s \tag{13}$$

Since the propagation delays are not necessarily integer multiples of the sampling interval, a fractional-delay interpolation method is used to construct the discrete channel.

The discrete channel is approximated by:

$$h[n] = \sum_{\ell=0}^{L-1} a_\ell w_\ell[n] \tag{14}$$

where $w_\ell[n]$ represents the finite-support windowed-sinc interpolation kernel associated with path ℓ.

The maximum delay window is 600 ns. Therefore, the number of channel samples is:

$$N_h = \left\lceil \frac{600 \text{ ns}}{10 \text{ ns}} \right\rceil \tag{15}$$

$$= 60 \tag{16}$$

The resulting discrete channel contains 60 samples.

3.3 OFDM frequency-domain channel estimation

The estimated frequency response is obtained from the received and transmitted OFDM symbols:

$$\widehat{H}[k] = \frac{Y[k]}{X[k]} \tag{17}$$

where $Y[k]$ is the received frequency-domain signal and $X[k]$ is the known transmitted QPSK symbol.

3.4 Conversion to the delay domain

The delay-domain representation is obtained by applying an inverse discrete Fourier transform:

$$\widehat{h}[n] = \frac{1}{N} \sum_{k=0}^{N-1} \widehat{H}[k] e^{j2\pi kn/N} \tag{18}$$

where $N = 1024$ is the FFT size.

The delay corresponding to sample n is:

$$\tau_n = \frac{n}{f_s} \tag{19}$$

The delay sample spacing is therefore:

$$\Delta\tau = \frac{1}{f_s} = 10 \text{ ns} \tag{20}$$

The total observation window is:

$$T_{\text{window}} = \frac{N}{f_s} \tag{21}$$

$$= \frac{1024}{100 \times 10^6} \tag{22}$$

$$= 10.24 \ \mu\text{s} \tag{23}$$

3.5 Magnitude normalization

The delay-profile magnitude is normalized with respect to the maximum magnitude of the reference profile:

$$h_{\text{norm}}[n] = \frac{|\widehat{h}[n]|}{\max_n |h_{\text{ref}}[n]|} \tag{24}$$

The magnitude in decibels is:

$$h_{\text{dB}}[n] = 20 \log_{10}(h_{\text{norm}}[n]) \tag{25}$$

A small positive numerical floor is used in the implementation to avoid taking the logarithm of zero.

3.6 Local peak detection

A sample is identified as a local maximum when its magnitude is greater than its neighbouring samples and exceeds the selected detection threshold.

The threshold used in this experiment is:

$$P_{\text{threshold}} = -25 \text{ dB} \tag{26}$$

The detected local maxima represent candidate delay-domain components. They do not necessarily correspond one-to-one with individual physical rays.

4 Ray-Tracing Results

The ray-tracing simulation produced twelve valid propagation entries.

Table 2 lists the propagation delays.

Table 2: Propagation delays obtained from Sionna RT

Path	Delay (ns)	Fractional sample index
1	133.426	13.343
2	134.112	13.411
3	145.273	14.527
4	147.918	14.792
5	180.333	18.033
6	180.333	18.033
7	180.842	18.084
8	180.842	18.084
9	223.079	22.308
10	228.234	22.823
11	277.659	27.766
12	283.282	28.328

5 Experimental Results

5.1 Detected delay-domain peaks

The peak detection algorithm identified five prominent local maxima.

The detected results are presented in Table 3.

Table 3: Detected delay-domain peaks

Peak	Estimated delay (ns)	Magnitude (dB)
1	130	-4.271
2	150	0.000
3	180	-16.015
4	230	-24.807
5	280	-22.940

5.2 Graphical results

Figure 1 shows the reference and estimated delay profiles.

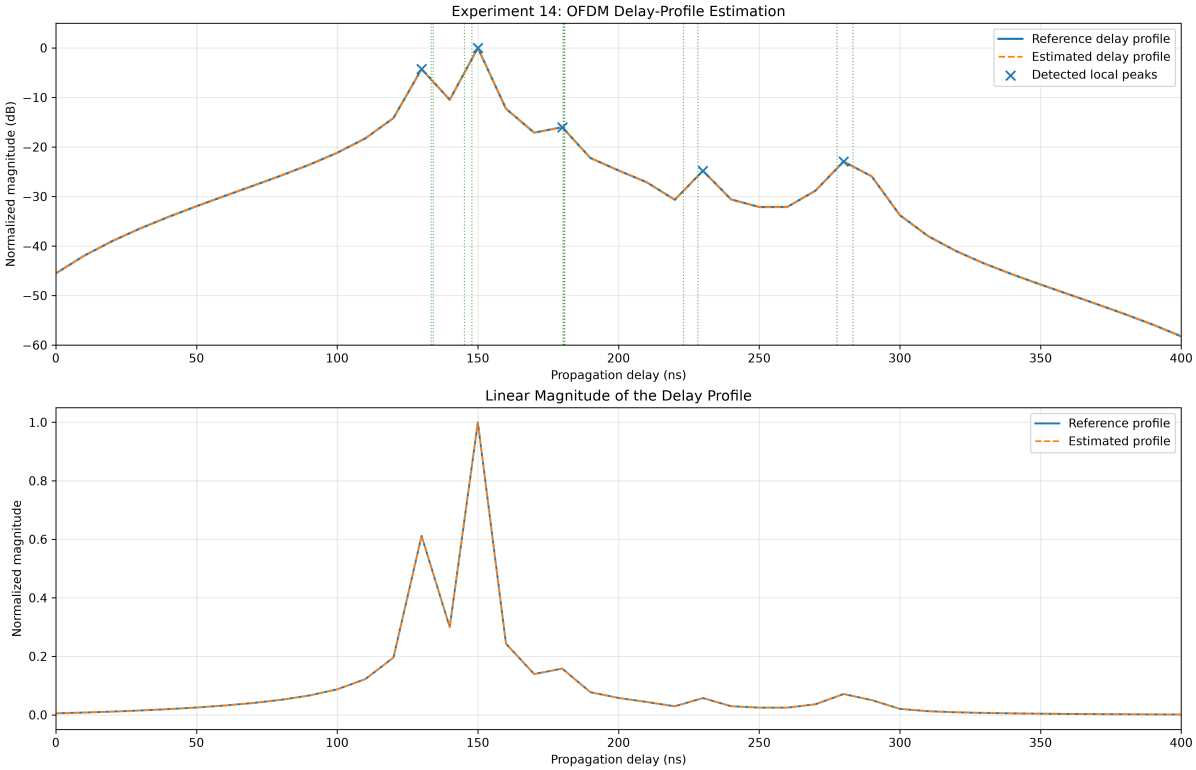


Figure 1: OFDM delay-profile estimation. The upper panel presents the normalized delay profile in decibels, including Sionna RT ray delays and detected local maxima. The lower panel presents the normalized linear magnitude profiles.

5.3 Description of the results

Upper panel: Delay profile in decibels

The upper panel compares the reference delay profile with the estimated delay profile.

The blue solid curve represents the reference profile, while the orange dashed curve represents the estimated profile. The green vertical lines indicate the individual propagation delays obtained from Sionna RT. The cross markers represent detected local maxima.

The estimated and reference curves overlap closely across the displayed delay interval.

Five prominent peaks are detected at approximately 130 ns, 150 ns, 180 ns, 230 ns, and 280 ns.

The strongest detected component occurs at 150 ns, with a normalized magnitude of 0 dB.

Lower panel: Linear magnitude

The lower panel presents the same profiles in linear magnitude.

The strongest component is observed around 150 ns, followed by a secondary component around 130 ns.

Weaker components are visible around 180 ns, 230 ns, and 280 ns.

The tails around the peaks are influenced partly by the finite-bandwidth response and fractional-delay interpolation. They should not automatically be interpreted as additional physical propagation paths.

6 Comparison of Ray Delays and Detected Peaks

The detected delay peaks can be approximately associated with groups of nearby Sionna RT ray delays.

Table 4: Approximate association of ray-delay groups with detected peaks

Detected group	Associated ray delays (ns)	Peak (ns)
1	133.426, 134.112	130
2	145.273, 147.918	150
3	180.333, 180.333, 180.842, 180.842	180
4	223.079, 228.234	230
5	277.659, 283.282	280

These are approximate associations based on delay proximity. They do not establish that each detected maximum uniquely identifies an individual physical ray.

7 Reconstruction Validation

The reference delay profile is calculated as:

$$h_{\text{ref}}[n] = \text{IFFT}\{H_{\text{ref}}[k]\} \tag{27}$$

The reconstruction error is:

$$e[n] = \widehat{h}[n] - h[n] \tag{28}$$

The error RMS is:

$$E_{\mathrm{RMS}} = \sqrt{\frac{1}{N_h} \sum_{n=0}^{N_h-1} |e[n]|^2} \tag{29}$$

The relative reconstruction error is:

$$\mathrm{RMSE}_{\mathrm{rel}} = \frac{E_{\mathrm{RMS}}}{H_{\mathrm{RMS}}} \tag{30}$$

The measured values are:

$$H_{\mathrm{RMS}} = 1.465363 \times 10^{-5} \tag{31}$$

$$E_{\mathrm{RMS}} = 2.691910 \times 10^{-21} \tag{32}$$

Therefore:

$$\boxed{\mathrm{RMSE}_{\mathrm{rel}} = 1.837026 \times 10^{-16}} \tag{33}$$

This value indicates near-identical reconstruction of the sampled channel under the noise-free simulation conditions.

8 Discussion and Research Findings

The experiment demonstrates that the OFDM frequency-domain channel estimate can be transformed into a delay-domain representation that closely reproduces the reference discrete channel.

The principal findings are:

1. **Delay-domain reconstruction:** The IFFT of the estimated frequency response successfully reproduces the reference delay profile.

2. **Multipath identification:** Twelve valid ray entries were obtained from Sionna RT, while five prominent local maxima were identified using the selected peak detection threshold.

3. **Closely spaced propagation components:** Several propagation paths have very small delay differences. For example:

$$134.112 - 133.426 = 0.686 \text{ ns} \tag{34}$$

   Such closely spaced paths are not independently distinguishable in the present configuration.

4. **Dominant delay components:** The strongest detected component occurs around 150 ns. Additional components are detected around 130 ns, 180 ns, 230 ns, and 280 ns.

5. **Numerical consistency:** The relative reconstruction RMSE of 1.837026×10^{-16} confirms near-identical reconstruction of the sampled channel.

6. **Finite bandwidth limitation:** The nominal delay scale associated with a 100 MHz bandwidth is approximately:

$$\Delta\tau \sim \frac{1}{B} = 10 \text{ ns} \tag{35}$$

This is a bandwidth-based resolution scale, not a guarantee of resolving two paths separated by exactly 10 ns.

9 Limitations

The following limitations apply to this experiment:

1. Additive noise and interference are disabled.
2. The propagation channel is static.
3. Doppler effects due to vehicle motion are not included.
4. The finite bandwidth limits the separation of closely spaced propagation components.
5. The number of detected peaks depends on the selected detection threshold.
6. Fractional-delay interpolation introduces approximation and boundary effects.
7. The estimated profile is compared against the same discrete channel used to generate the received signal. This is an internal consistency check rather than independent physical validation.
8. No direct target range, velocity, or position estimation is performed.

10 Relevance to V2X-ISAC Research

This experiment establishes a baseline for delay-domain processing in subsequent V2X Integrated Sensing and Communication research.

The nominal delay-resolution scale associated with bandwidth B is:

$$\Delta\tau \sim \frac{1}{B} \tag{36}$$

For a conventional monostatic radar, the corresponding nominal range-resolution scale is:

$$\Delta R \sim \frac{c}{2B} \tag{37}$$

For an illustrative bandwidth of 100 MHz:

$$\Delta R \sim \frac{299792458}{2(100 \times 10^6)} \tag{38}$$

$$\approx 1.50 \text{ m} \tag{39}$$

This is a theoretical bandwidth-based resolution estimate, not a measured result from this experiment.

The current simulation uses spatially separated transmitter and receiver locations. Consequently, propagation delays must be interpreted using bistatic geometry rather than applying the monostatic range equation directly.

11 Conclusion

Experiment 14 successfully demonstrated OFDM-based delay-profile estimation using a multipath channel derived from Sionna RT at 5.9 GHz.

The estimated frequency response was transformed into the delay domain using a 1024-point IFFT.

The resulting profile closely matched the reference discrete channel, with a relative reconstruction RMSE of 1.837026×10^{-16}.

Although Sionna RT produced twelve valid ray entries, the peak detection algorithm identified five prominent delay-domain maxima. This demonstrates that multiple propagation components can contribute to a common delay-domain peak when their delays are closely spaced relative to the effective resolution of the waveform.

The experiment establishes a baseline for delay-domain processing while highlighting the limitations imposed by finite bandwidth and multipath overlap.

The next experiment will introduce additive white Gaussian noise at different Signal-to-Noise Ratio (SNR) levels to investigate the robustness of delay estimation and the detection of weaker propagation components.